

CHARLES K. GIFFORD

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EDUCATION

Boston College, Chestnut Hill, MA

Expected: May 2028

Bachelor of Science: Computer Science; Minor: Mathematics

GPA: 3.33

Honors: Red Bandanna Leadership Fellow (0.03% of students)

EMPLOYMENT

Data Analytics Intern, *The Port Authority of New York and New Jersey*, New York, NY May 2026 - Present

- Developed a Python pipeline to classify the language of over 93,000 multilingual toll-dispute complaint records from the E-ZPass case management system, layering statistical language detection (lingua) with quality gates that strip system-generated templates and low-signal noise; isolates clean, analyzable English records for an LLM to ingest and sort the complaints.
- Engineer an end-to-end Python data pipeline to cross-reference 2,200+ fleet vehicles against an Oracle SQL database, identifying ~31% of vehicles out of compliance with over \$95,000 in outstanding tolls.
- Build interactive geospatial tools (Python, Folium, GeoJSON) with custom JavaScript cross-filtering, heatmaps, and roadside-image lookups to investigate transponder misuse, surfacing fleet tags transacting well outside expected operations (as far as Illinois and Florida); tools adopted by the Revenue Recovery team for a separate enforcement effort.

Teaching Assistant, *Boston College Computer Science Department*, Chestnut Hill, MA August 2025 - Present

- Assist professor in teaching the programming fundamentals of Python to more than 50 students by leading weekly discussion sections and holding daily office hours.
 - Review and grade assignments and provide constructive, positive feedback on efficiency, readability, and precision, with the hopes of strengthening their problem-solving skills and algorithmic thinking ability.
 - Collaborate with the professor to improve course content and create supplemental coding exercises.
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EXTRACURRICULAR

Student Leader & Big Brother, *BC Bigs*, Chestnut Hill, MA

October 2024 - Present

- Lead biweekly family meetings and coordinate events such as "Meet the Bigs Day" to foster community and strengthen connections between Bigs and Littles across partner schools,
 - Mentor a Boston Renaissance Charter School student through more than 50 weekly meetings, designing activities that build academic engagement, teamwork, and personal development.
 - Maintain communication with Big Brothers Big Sisters staff and school supervisors to track match progression and ensure program success.
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PROJECTS

Co-Founder, *W2WSports*, W2W-Sports.com

October 2025 - Present

- Engineered a proprietary multi-factor watchability scoring algorithm across 10+ sports leagues, weighting variables such as rivalry, marketability, competitiveness, and game importance to rank live sporting events in real time.
 - Built a full-stack Flask web application integrating the ESPN API to ingest, process, and score live game data, serving ranked results dynamically to users across multiple sport verticals.
 - Developed an automated email digest system using Resend and Supabase, deploying Vercel Cron jobs to deliver personalized sport rankings to subscribers on a scheduled cadence.
 - Implemented a Pillow-based Open Graph share card generator to produce dynamic image previews per game, increasing content shareability and social media engagement potential.
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TECHNICAL SKILLS

- Python (Pandas, Numpy), SQL, Java, JavaScript C, JavaScript, Flask, React, FastAPI, Git, Oracle, HTML, CSS